MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, ECL, FLIP-FLOPS, MONOLITHIC SILICON

This amendment forms a part of MIL-M-38510/61B,
dated 15 July 1985, and is approved for used by
all Departments and Agencies of the Department of Defense.

PAGE 18

Figure 8, after "t_p (clock) = 40 ns", insert "PRR = 1 MHz"

PAGE 21

Figure 10, after "Q OUTPUT" waveform and before "GENERATOR CHARACTERISTICS", insert:

$$\text{FREQ. (OUT)} = \frac{\text{FREQ. (IN)}}{2}$$

FMAX IS HIGHEST INPUT FREQUENCY AT WHICH DEVICE CEASES TO TOGGLE

PAGE 32

TABLE III, F_{MAX} row, Min column of subgroup 9, delete " 62.5" and substitute "125" in two places.
TABLE III, F_{MAX} row, Min column of subgroup 10, delete " 62.5" and substitute "125" in two places.
TABLE III, F_{MAX} row, Min column of subgroup 11, delete " 52.5" and substitute "105" in two places.

PAGE 34

TABLE III, I_{IH3}, Max column of subgroup 2, delete "420" and substitute "375" in two places.
TABLE III, I_{IH4}, Max column of subgroup 2, delete "450" and substitute "495" .

PAGE 35

TABLE III, F_{MAX} row, Min column of subgroup 9, delete "100" and substitute "200" in two places.
TABLE III, F_{MAX} row, Min column of subgroup 10, delete "100" and substitute "200" in two places.
TABLE III, F_{MAX} row, Min column of subgroup 11, delete " 100" and substitute "200" in two places.

PAGE 37

TABLE III, F_{MAX} row, Min column of subgroup 1, delete "62.5" and substitute "125".
TABLE III, F_{MAX} row, Min column of subgroup 2, delete "62.5" and substitute "125".
TABLE III, F_{MAX} row, Min column of subgroup 3, delete " 57.5" and substitute "115".

PAGE 43

TABLE III, F_{MAX} row, Min column of subgroup 9, delete "62.5" and substitute "125" in two places.
TABLE III, F_{MAX} row, Min column of subgroup 10, delete "57.5" and substitute "115" in two places.
TABLE III, F_{MAX} row, Min column of subgroup 11, delete "52.5" and substitute "125" in two places.

The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Page replaced
14	new page
16	new page
17	new page

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC, SH, OS
Air Force - 11, 19, 85, 99

Preparing activity:
DLA - CC

(Project 5962-1791)

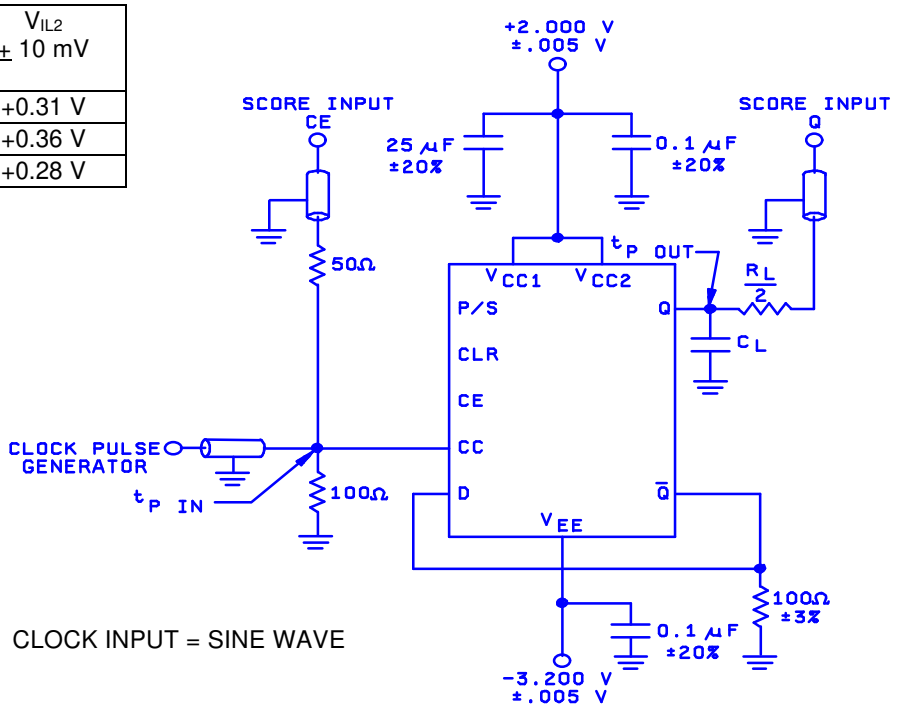
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GENERATORS CHARACTERISTICS

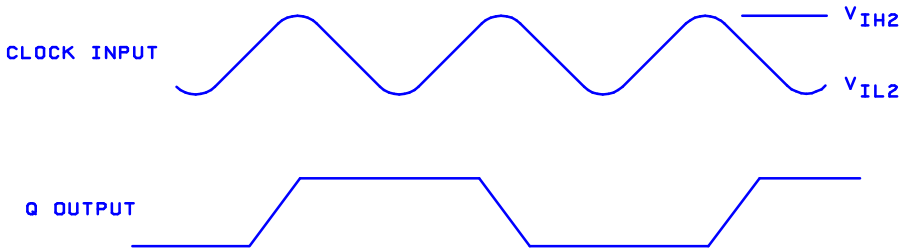
T_C	V_{IH2} ± 10 mV	V_{IL2} ± 10 mV
25 °C	+1.11 V	+0.31 V
125 °C	+1.24 V	+0.36 V
-55 °C	+1.01 V	+0.28 V

$\frac{R_L}{2} = 50\Omega \pm 5\%$

SCOPE INPUT = 50Ω to GND
C_L (test jig) ≤ 5 pF



CLOCK INPUT = SINE WAVE



FREQUENCY (OUT) = $\frac{\text{FREQUENCY (IN)}}{2}$

FMAX IS HIGHEST INPUT FREQUENCY AT WHICH DEVICE CEASES TO TOGGLE

NOTES:
1. Perform test in accordance with test table; each output is tested separately.
2. All input and output cables are equal lengths of 50 ohm coaxial cables. Wire length should be ≤ .250 (6.35 mm) from t_p in to input pin and t_p out to output pin.
3. Outputs not under test connected to a 100 ohm resistor to ground.
4. Note that observed pulse amplitude is attenuated by one half.

FIGURE 4. F_{MAX} test circuit for device types 01 and 02.

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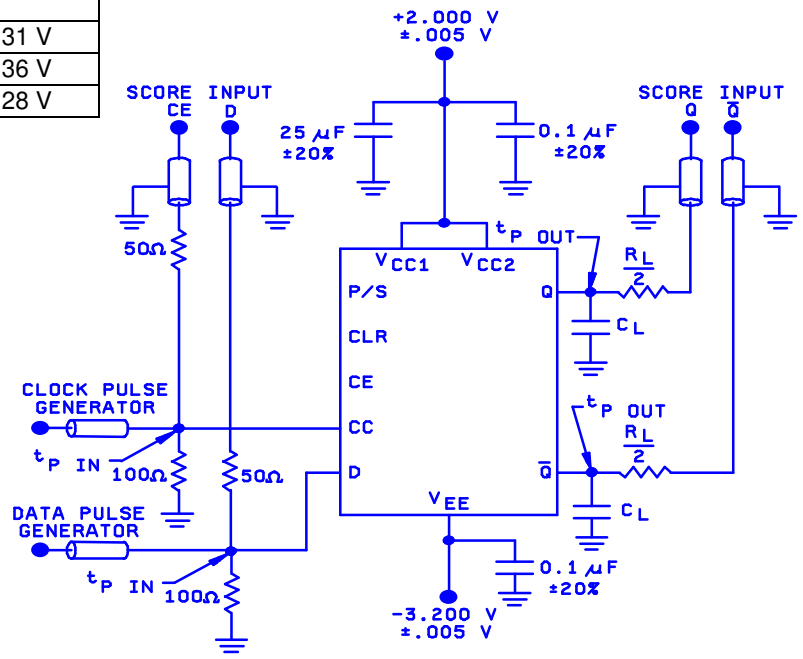
GENERATORS CHARACTERISTICS

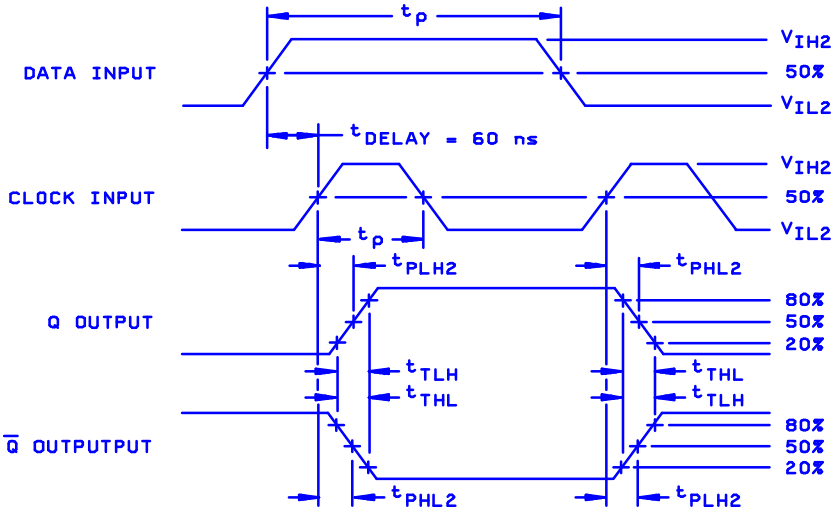
T_C	V_{IH2} ± 10 mV	V_{IL2} ± 10 mV
25 °C	+1.11 V	+0.31 V
125 °C	+1.24 V	+0.36 V
-55 °C	+1.01 V	+0.28 V

Z_{OUT} = 50Ω
t_p (data) = 150 ns
t_p (clock) = 40 ns
PRR = 1 MHz
Device type 01
T_{THL} = 2.0 ns (20%-80%)
T_{TLH} = 2.0 ns (20%-80%)
Device type 02
T_{THL} = 1.5 ns (20%-80%)
T_{TLH} = 1.5 ns (20%-80%)

$\frac{R_L}{2}$ = 50Ω ± 5%

SCOPE INPUT = 50Ω to GND
C_L (test jig) ≤ 5 pF





NOTES:
1. Perform test in accordance with test table; each output is tested separately.
2. All input and output cables are equal lengths of 50 ohm coaxial cables. Wire length should be ≤ .250 (6.35 mm) from t_p in to input pin and t_p out to output pin.
3. Outputs not under test connected to a 100 ohm resistor to ground.
4. Note that observed pulse amplitude is attenuated by one half.

FIGURE 6. Synchronous switching test circuit for device types 01 and 02.

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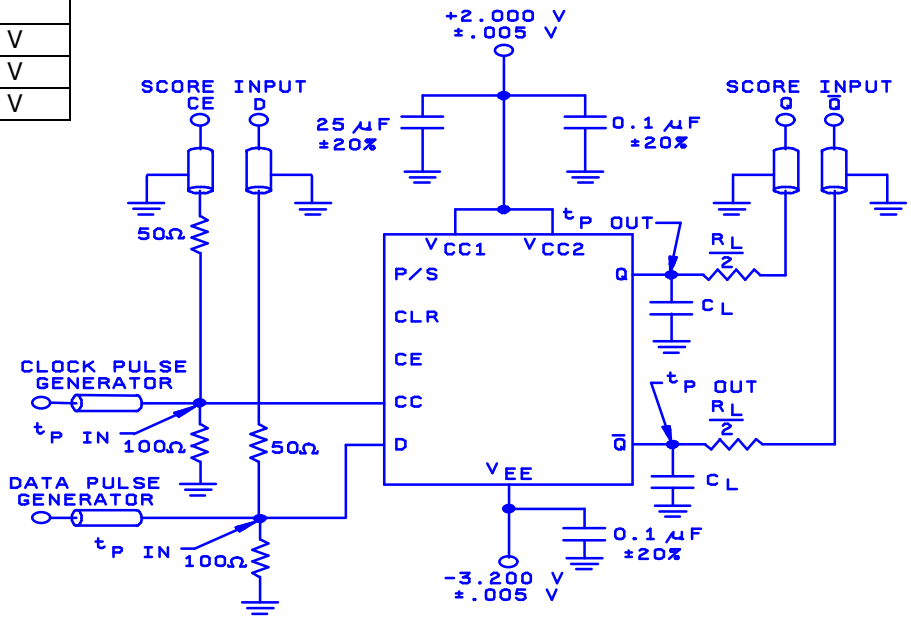
GENERATORS CHARACTERISTICS

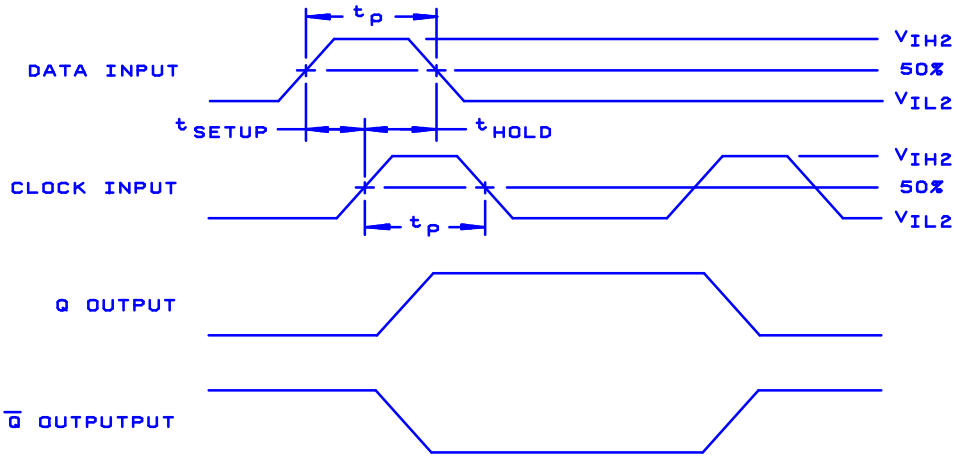
T_C	V_{IH2} ± 10 mV	V_{IL2} ± 10 mV
25 °C	+1.11 V	+0.31 V
125 °C	+1.24 V	+0.36 V
-55 °C	+1.01 V	+0.28 V

Z_{OUT} = 50Ω
t_p (data) = 40 ns
t_p (clock) = 40 ns
PRR = 1 MHz
Device type 01
T_{THL} = 2.0 ns (20%-80%)
T_{TLH} = 2.0 ns (20%-80%)
Device type 02
T_{THL} = 1.5 ns (20%-80%)
T_{TLH} = 1.5 ns (20%-80%)

$\frac{R_L}{2}$ = 50Ω ± 5%

SCOPE INPUT = 50Ω to GND
C_L (test jig) ≤ 5 pF





NOTES:
1. Perform test in accordance with test table; each output is tested separately.
2. All input and output cables are equal lengths of 50 ohm coaxial cables. Wire length should be ≤ .250 (6.35 mm) from t_p in to input pin and t_p out to output pin.
3. Outputs not under test connected to a 100 ohm resistor to ground.
4. Note that observed pulse amplitude is attenuated by one half.

FIGURE 7. Setup and hold test circuit for device types 01 and 02.

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